Inventory Management

Assignment 2025–2026

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This report reproduces the structure of the original submission, with the calculations re-derived and corrected. All figures match a fully formula-driven spreadsheet (`ERGASIA_EN.xlsx`); every result is a live computation, not a hard-coded number.

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1. Part A — Single-item deterministic model

A.1 Demand forecasting (simple exponential smoothing)

We must choose the smoothing constant that gives the most accurate forecasts over months 13–24, among $\alpha \in \{0.10, 0.15, 0.20\}$. The initial estimate is the mean demand of the first 11 months:

$$\hat{a}_{11} = \frac{1}{11}\sum_{i=1}^{11} x_i = 200.64 \text{ units.}$$

Simple exponential smoothing updates the forecast through

$$\hat{a}_t = \hat{a}_{t-1} + \alpha\,(x_t - \hat{a}_{t-1}),$$

where $\hat{a}_t$ is the forecast issued after month t for month $t+1$, and x_t is the actual demand in month t. For each α the forecast series is generated recursively; the one-step error is $e_t = x_t - \hat{a}_t$, and accuracy over months 13–24 is measured with

$$\text{MSE} = \tfrac{1}{n}\sum e_t^2, \qquad \text{MAD} = \tfrac{1}{n}\sum |e_t|, \qquad \text{MAPE} = \tfrac{1}{n}\sum \left|\tfrac{e_t}{x_t}\right|.$$

Table 1: Error measures over months 13–24.

	$\alpha = 0.10$	$\alpha = 0.15$	$\alpha = 0.20$
MSE	1357.4	1270.0	**1176.0**
MAD	32.78	31.68	**30.45**
MAPE	0.1880	0.1822	**0.1754**

All three measures are minimized at the same value, so we select $\alpha = 0.20$. Since no further data exist, the forecast stays constant for every month 25–36 at the value of $\hat{a}_{24}$:

$$\boxed{\hat{a}_{24} = 189.6 \approx 190 \text{ units}}.$$

If new actual demand x_t became available, the forecast would be revised.

A.2 Reorder point and Economic Order Quantity

Treating monthly demand as constant at the forecast level, 190 units/month, the annual demand is

$$D = 12 \times 190 = 2280 \text{ units/year.}$$

With lead time $L = 1$ month, the reorder point equals the demand over the lead time:

$$s = L \cdot (\text{monthly demand}) = 190 \text{ units.}$$

Minimizing the total cost $TC = Dv + \frac{D}{Q}A + vr\frac{Q}{2}$ with respect to Q gives the EOQ; with $A = 750$, $v = 100$, $r = 0.10$,

$$Q^* = \text{EOQ} = \sqrt{\frac{2AD}{vr}} = 584.8 \xrightarrow{\text{round to mult. of 10}} 580 \text{ units.}$$

The annual ordering and holding costs at $Q^* = 580$ are

$$C_{\text{ord}} = \frac{D}{Q}A = 2948.3/\text{yr}, \qquad C_{\text{hold}} = vr\frac{Q}{2} = 2900/\text{yr},$$

$$\boxed{TC = \frac{D}{Q}A + vr\frac{Q}{2} = 5848.3/\text{yr}}.$$

A.3 All-units quantity discounts

The supplier offers $d_1 = 1\%$ for $Q \geq 2000$ and $d_2 = 2\%$ for $Q \geq 5000$. Under an all-units scheme the EOQ at the discounted holding cost is $\mathrm{EOQ}_d = \sqrt{2AD/\big(v(1-d)r\big)}$, giving $\mathrm{EOQ}_{d_1} = 587.8$ and $\mathrm{EOQ}_{d_2} = 590.7$ — both far below their thresholds. The discount can therefore only be obtained by ordering *at* the threshold Q_b, so we compare the full annual cost (purchase + ordering + holding):

Table 2: Total annual cost including purchase cost $D\,v(1-d)$.

Scenario	Q	Purchase	Ordering	Holding
Q^* (no discount)	580	228 000	2948.3	2900
Q_{b1}, $d_1 = 1\%$	2000	225 720	855	9900
Q_{b2}, $d_2 = 2\%$	5000	223 440	342	24 500

$$TC(Q^*) = \mathbf{233\,848}\ , \quad TC(Q_{b1}) = 236\,475\ , \quad TC(Q_{b2}) = 248\,282\ .$$

The lowest total cost occurs at $Q^* = 580$ (no discount): the savings on the purchase price do not offset the sharp rise in holding cost caused by the very large order quantities. **Accepting either discount is not worthwhile**; it is cheaper to place more frequent, smaller orders.

2. Part B — Stochastic demand (transformers)

Weekly demand is normal, $X \sim N(\mu, \sigma^2)$ with $\mu = 140$, $\sigma = 40$ units/week, and lead time $L = 2$ weeks. Over the lead time,

$$\mu_L = L\mu = 280, \qquad \sigma_L = \sqrt{L}\,\sigma = 56.57, \qquad D = 52\mu = 7280 \text{ units/year.}$$

The common order quantity is the EOQ (with $A = 100$, $v = 80$, $r = 0.10$):

$$Q = \mathrm{EOQ} = \sqrt{\frac{2AD}{vr}} = 426.6 \approx 427 \text{ units.}$$

B.1 Two continuous-review (s, Q) systems

System 1 — Department's choice (target fill rate $P_2 = 95\%$). From $P_2 = 1 - \sigma_L G(k)/Q$ we need

$$G(k) = \frac{Q(1-P_2)}{\sigma_L} = 0.377 \ \Rightarrow\ k = 0.04, \quad ss = k\sigma_L \approx 2, \quad s = \mu_L + ss = 282.$$

The per-cycle no-stockout probability is $P_1 = \Phi(k) = 51.6\%$, so $(s, Q) = (282, 427)$.

System 2 — General Manager's wish ($P_1 = 95\%$). Here $P_1 = 1 - \frac{1}{20} = 0.95 = \Phi(k) \Rightarrow k = 1.645$, hence

$$ss = k\sigma_L = 93, \quad s = \mu_L + ss = 373, \quad P_2 = 1 - \frac{\sigma_L G(k)}{Q} = 99.7\%,$$

i.e. $(s, Q) = (373, 427)$. The very high service comes from a large safety stock.

B.2 Cost comparison and defense of the choice

Each system's annual cost combines ordering, holding and shortage terms; the shortage term uses the uniform convention $C_{\text{short}} = B\,(1-\Phi(k))\,D/Q$, with the per-unit back-order discount $B = 5\$$.

Table 3: Annual cost breakdown of the two (s, Q) systems.

Cost component	System 1 (Dept)	System 2 (GM)
Average inventory $ss + Q/2$	215.5	306.5
Ordering $(D/Q)A$	1704.9	1704.9
Holding $vr(ss + Q/2)$	1724	2452
Shortage $B(1 - P_1)(D/Q)$	41.3	4.3
Total TC	**3470.2**	**4161.2**

System 1 holds far less safety stock and is the cheaper system. Over the requested 18 months (1.5 years) the Department's choice saves

$$(TC_2 - TC_1) \times 1.5 = (4161.2 - 3470.2) \times 1.5 = \mathbf{1036}\ .$$

The Department's lower-service ($P_2 = 95\%$) policy did *not* cost the company; it saved money while keeping a reasonable service level with much less inventory.

B.3 Improved (s, Q) with explicit shortage cost

With a shortage cost $B = 5$/unit, the optimal reorder point satisfies the joint optimality condition (the per-cycle stockout probability depends on Q):

$$1 - \Phi(k) = \frac{Q\,v\,r}{B\,D} = \frac{427 \cdot 80 \cdot 0.1}{5 \cdot 7280} = 0.0938 \;\Rightarrow\; \Phi(k) = 0.9062 \;\Rightarrow\; k = 1.317.$$

Keeping $Q = \text{EOQ} = 427$, this gives

$$ss = k\sigma_L = 75, \qquad s = \mu_L + ss = 355, \qquad P_2 = 1 - \frac{\sigma_L G(k)}{Q} = 99.4\%.$$

Table 4: Comparison of the three (s, Q) systems.

	System 1 (Dept)	System 2 (GM)	System 3 (Improved)
s	282	373	355
ss	2	93	75
Average inventory	215.5	306.5	288.5
P_2	0.950	0.997	0.994
Ordering $(D/Q)A$	1704.9	1704.9	1704.9
Holding $vr(ss + Q/2)$	1724	2452	2308
Shortage $B(1 - \Phi(k))D/Q$	41.3	4.3	8.0
Total TC	**3470.2**	4161.2	4020.9

The improved System 3 substantially betters System 2 (much lower inventory and cost) while reaching a high fill rate ($\approx 99\%$), but it still does not beat System 1 on total cost. **System 1, $(s, Q) = (282, 427)$, remains the cheapest.**

B.4 Periodic-review (R, S) systems and final recommendation

We now switch to periodic review with order-up-to level $S = \mu(R{+}L) + k\,\sigma(R{+}L)$, lead time $L = 2$ weeks, and two review periods $R_1 = 4$ weeks (once/month) and $R_2 = 2$ weeks (twice/month). For the protection interval $R + L$:

$$R_1 :\ \mu(R{+}L) = 840,\ \sigma(R{+}L) = 97.98; \qquad R_2 :\ \mu(R{+}L) = 560,\ \sigma(R{+}L) = 80.$$

B.4(a) — target $P_1 = 95\%$. $\Phi(k) = 0.95 \Rightarrow k = 1.645$, hence

$$S_1 = 840 + 1.645(97.98) = 1001, \qquad S_2 = 560 + 1.645(80) = 692,$$

with mean time between stockouts $TBS = R/(1 - P_1)$ equal to 80 and 40 weeks respectively.

B.4(b) — minimum-cost design. The cost-minimizing service level uses the critical ratio

$$CR = \frac{B}{B + v\,r\,R} \quad [R \text{ in years}] \;\Rightarrow\; k = \Phi^{-1}(CR).$$

$$R_1: \; CR = \frac{5}{5 + 80(0.1)(4/52)} = 0.890 \Rightarrow k = 1.229,\; S = 960,\; P_2 = 99.1\%,$$

$$R_2: \; CR = \frac{5}{5 + 80(0.1)(2/52)} = 0.942 \Rightarrow k = 1.572,\; S = 686,\; P_2 = 99.3\%.$$

B.4(c) — final comparison and recommendation. The supplier offers a 3% purchase-price discount *only* if the company commits to fixed review periods, i.e. only for the periodic (R, S) systems. The discounted price is $v' = v(1 - 0.03) = 77.6$ /unit, so the annual purchase cost for any (R, S) system is $D\,v' = 564\,928$, while the continuous (s, Q) system pays the full $D\,v = 582\,400$.

Table 5: Total annual cost (purchase + operating). The 3% discount applies only to the periodic systems.

System	S / type	P_2	Total cost ()
(R, S), R_1=4, P_1=95%	1001	0.996	569 655
(R, S), R_2=2, P_1=95%	692	0.994	569 642
(R, S), R_1=4, min cost	960	0.991	**569 342**
(R, S), R_2=2, min cost	686	0.993	569 598
(s, Q) improved (full price)	355	0.994	586 421

Every periodic (R, S) system with the 3% discount ($\approx 569\,000$) is clearly cheaper than the continuous (s, Q) at full price ($\approx 586\,000$): the purchase-price saving of about 17 000/year dwarfs the differences in ordering, holding and shortage costs. The cheapest option overall is the once-a-month, minimum-cost design ($R_1 = 4$ weeks, $S = 960$).

Recommendation. Accept the supplier's fixed-period proposal and order *once per month* ($R_1 = 4$ weeks). This captures the 3% purchase discount, which makes it the lowest-total-cost policy ($\approx 569\,342$/year) while still delivering a high fill rate ($P_2 \approx 99\%$).

References

Nenes, G. (2025). *Inventory and Supply Chain Management.* Propobos.
Ioannou, G. (2019). *Production and Service Operations Management.* Unibooks.